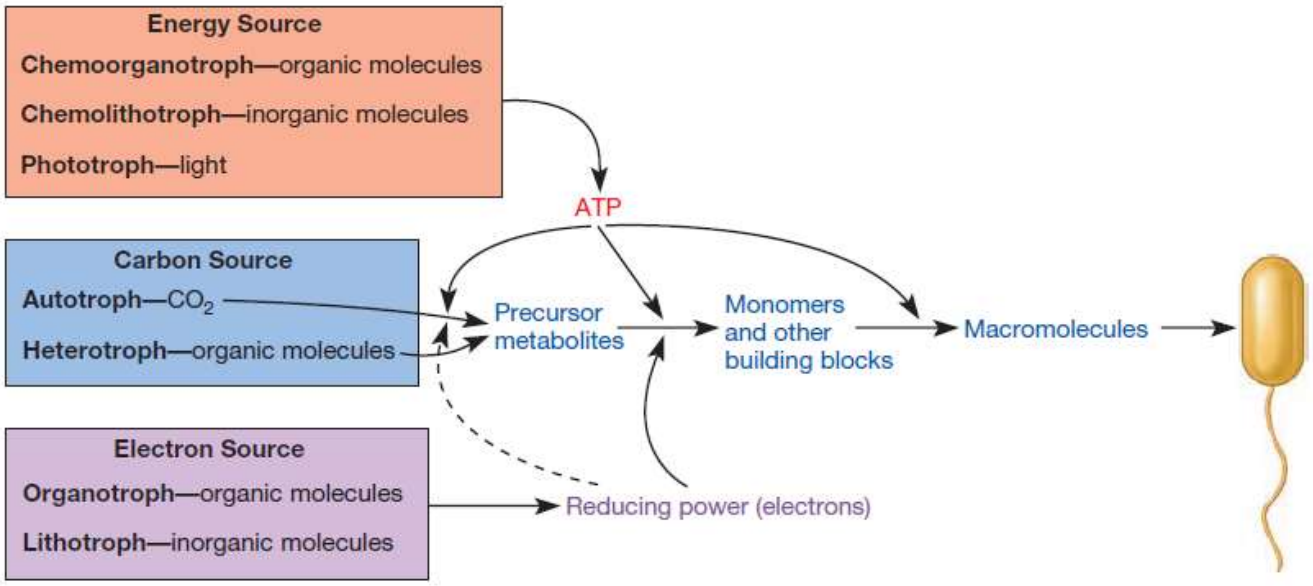
MICROBIAL METABOLISM

- Metabolism is the term used to describe all the **biochemical reactions** that take place inside a cell
- It includes those reactions that release **energy** into the cell, and those that make use of that energy.
- **Catabolism** describe reactions that **breakdown** large molecules, usually coupled to a release of energy.
- **Anabolism** describe reactions involved in the **synthesis** of macromolecules, usually requiring an input of energy
- As most microorganisms obtain their energy from the nutrients they take into the cell, that are from an organic or an inorganic source.
- Once inside the cell, these nutrients are processed by reactions that trap their chemical energy and breaking them into smaller molecules.
- These then serve as building blocks for the synthesis of new cellular components.
- Chemical compounds contain potential energy within their molecular structure, and some of this can be released when they are broken down.
- In other metabolic types, energy is obtained from the sun by means of **photosynthesis**.
- Central to the metabolic processes of any cell are **enzymes.**
- Without enzymes, many biochemical reactions simply wouldn’t take place at a rate fast enough for living cells to maintain themselves.



REQUIREMENT OF ENERGY

Like all other living things, microorganisms need to acquire energy in order to survive.

Energy is required:

- to **maintain** the structural integrity of the cell by repairing any damage to its constituents
- to **synthesize** new cellular components such as nucleic acids, polysaccharides and enzymes
- to **transport** certain substances into the cell from its surroundings for the cell to grow and multiply
- for cellular **movement.**

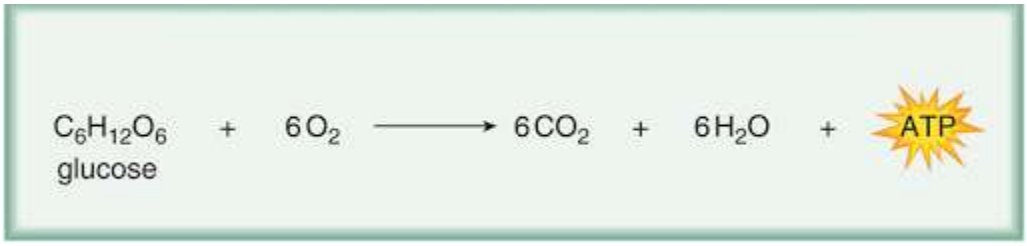
PRINCIPLE OF ENERGY GENERATION

- Energy taken up by the cell, be it in the form of nutrients or sunlight, must be converted into a usable form.
- The source from which most of the cells derive energy in cellular metabolism is called **adenosine triphosphate** (ATP).
- ATP is known as high-energy transfer compounds, which store the energy from the breakdown of nutrients and release it when required by the cell.

- In catabolic (breakdown) reactions, energy is released in the form of ATP, which can then be utilised in anabolic (synthetic) reactions.
- ATP has a structure very similar to the **nucleotides** found in RNA, except it has two additional phosphate groups
- When a bond is broken, large amount of energy is released, so when ATP is broken down to ADP, energy is made available to the cell.
- The process of adding or removing a phosphate group is called **phosphorylation** or **dephosphorylation**, respectively.

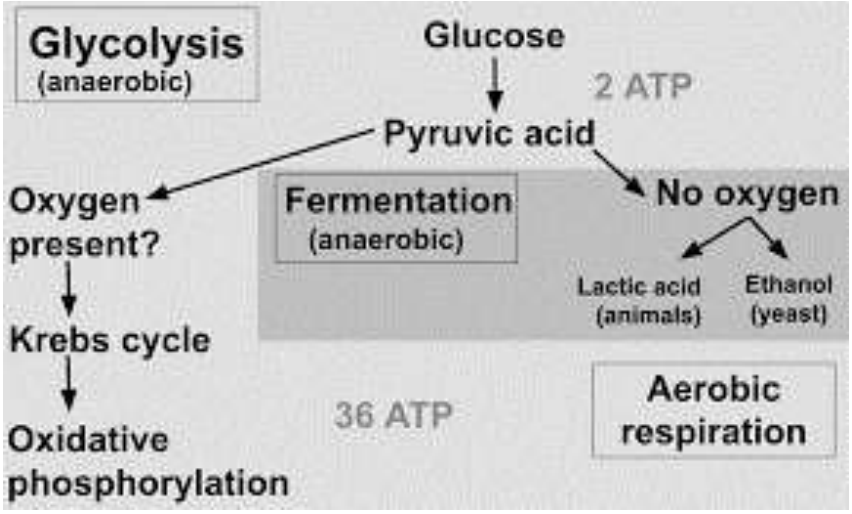
CARBOHYDRATE METABOLISM

- Carbohydrates are the chosen biomolecules when it comes to providing instant energy
- They are the first choice for generating energy in almost all the organisms
- Carbohydrates are available in the nature in the form of simple sugars like glucose or fructose or they can be present in the form of complex carbohydrates like the starch or glycogen
- While the simple carbohydrates can be used for generating energy directly, the complex carbohydrates need to be broken to simple sugars (via digestion) before they be utilized for generating energy
- Glucose is the most common simple sugar that is used universally by all organisms to generate energy
- When this glucose is completely oxidized to carbon dioxide and water energy in the form of ATP is obtained



- The above reaction happens if the organisms are aerobic and there is sufficient oxygen necessary for the oxidation process

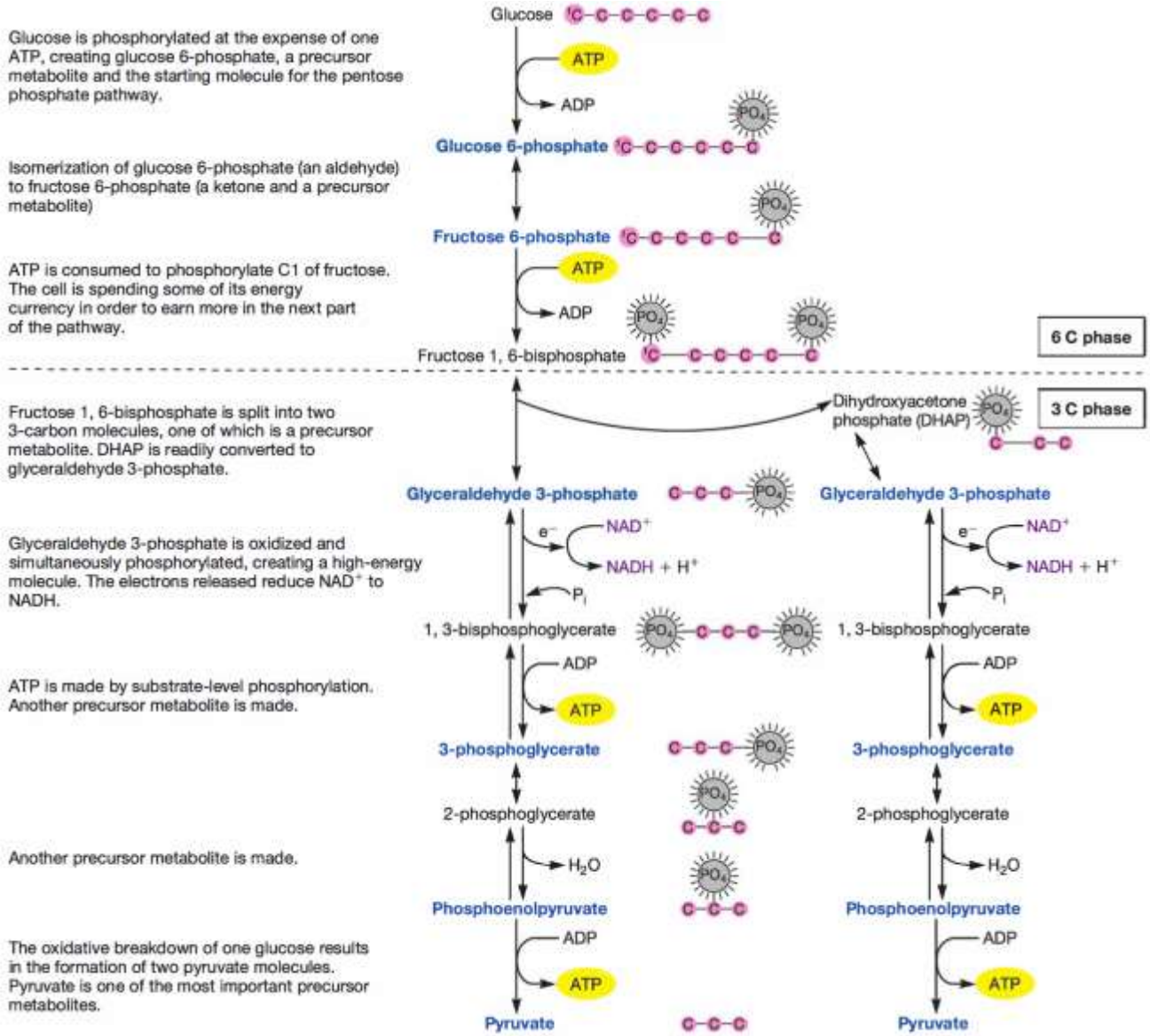
- In such organisms the glucose which is a six-carbon molecule is first broken down to 2 molecules of pyruvate (3 carbon molecule) by a process called glycolysis and this pathway does not require oxygen
- The subsequent steps where in pyruvate is converted to carbon dioxide requires oxygen and is called the Krebs's Cycle.
- Now, depending on the presence of oxygen the breakdown products of glucose (pyruvate) can take different pathways as summarized in the image given below



GLYCOLYSIS (EMBDEN-MEYERHOF PATHWAY)

- Carbohydrates are the chosen biomolecules when it comes to providing instant energy
- They are the first choice for generating energy in almost all the organisms
- Carbohydrates are available in the nature in the form of simple sugars like glucose or fructose or they can be present in the form of complex carbohydrates like the starch or glycogen
- While the simple carbohydrates can be used for generating energy directly, the complex carbohydrates need to be broken to simple sugars (via digestion) before they be utilized for generating energy
- Glucose is the most common simple sugar that is used universally by all organisms to generate energy

- When this glucose is completely oxidized to carbon dioxide and water, energy in the form of ATP is obtained
    - Location: cytoplasm of the cell
    - Oxygen requirement: occurs in the presence or absence of oxygen
- The Glycolysis process which can take place even under anaerobic conditions, the 6 carbon glucose molecule is broken down to 2 molecules of 3 carbon pyruvate through a series of reactions during which 2 NADH molecules and a net of 2 ATP molecules are formed

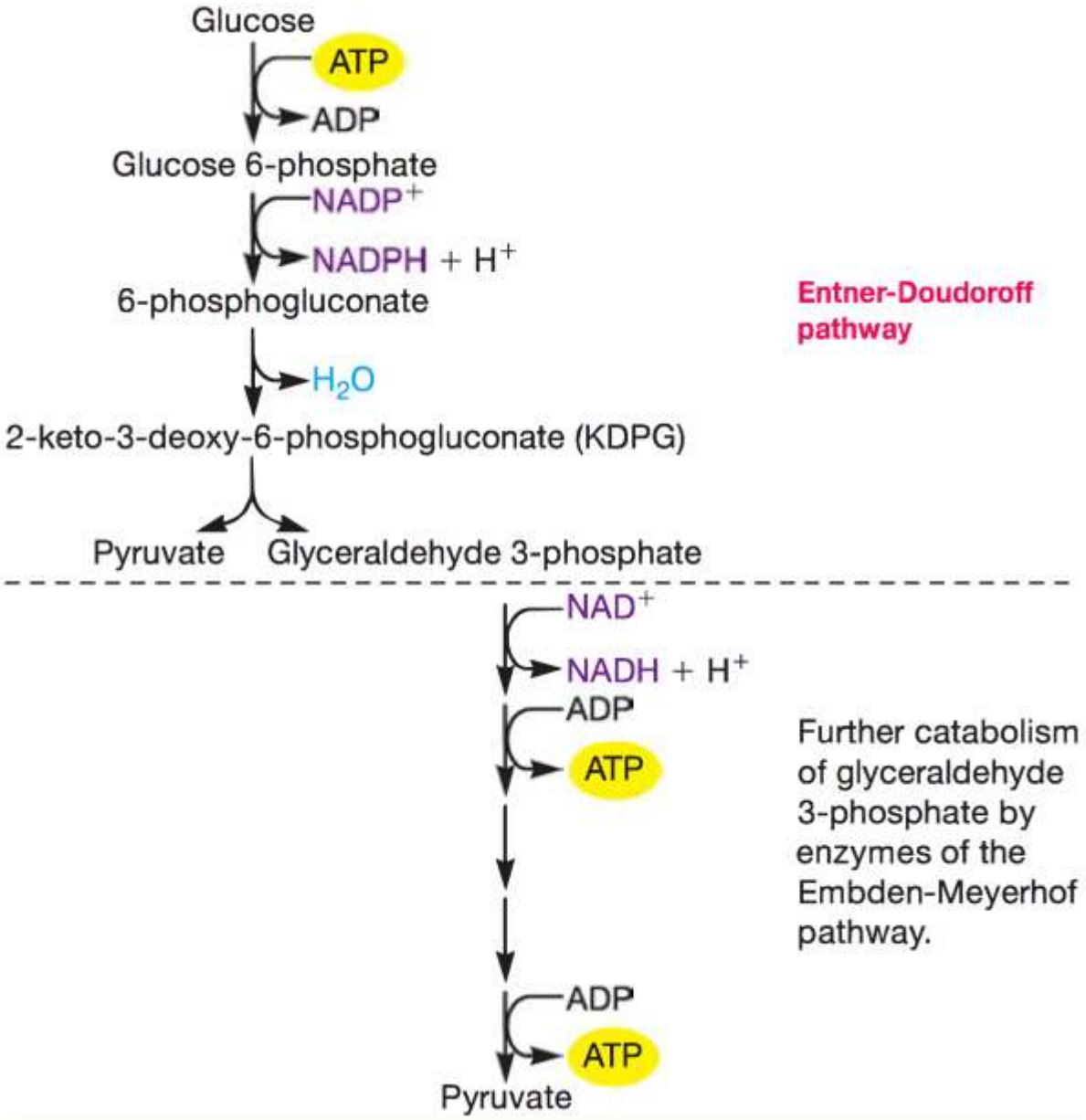


- Since these ATPs are formed without going through the oxidative phosphorylation process, they are said to be a result of substrate-level phosphorylation
- The overall reaction of glucose catabolism to pyruvate can be represented by the following equation

  Glucose + 2 ADP + 2 P_i + 2 NAD^+ → 2 pyruvate + 2 ATP + 2 $NADH^+$ + 2 H^+

ENTNER-DOUDOROFF PATHWAY

- While the Glycolysis also called the EMP pathway is the most common route used for the conversion of glucose to pyruvate, some soil bacteria like *Azobacter*, *Rhizobium* etc. use a slightly different pathway for breaking down glucose
- This pathway is named after the people who discovered it and is called the Entner-Doudoroff or ED pathway
- This pathway differs from the EMP pathway. Here, a pyruvate and a glyceraldehydes-3-phosphate molecule is formed during the initial breakdown of glucose
- This pathway has a key intermediate, 2-keto-3-deoxy-6-phosphoglucanate (KDPG), which is formed from glucose after going through 3 reactions and consuming 1 ATP
- This pathway uses NADP in the initial reactions to generate NADPH
- The glyceraldehyde-3-phosphate which is formed during the initial breakdown of glucose is catabolized to form a second pyruvate molecule using the same enzymes that are involved in the EMP pathway
- This pathway yields 1 ATP, 1 NADH and 1 NADPH
- While NADH is used to generate ATP via electron transport and oxidative phosphorylation, NADPH is used as reducing power for anabolic reactions



CITRIC ACID CYCLE (KREBS'S CYCLE/TRI CARBOXYLIC ACID CYCLE (TCA))

- The citric acid cycle is a cycle which more of a continuation to the glycolysis or the ED pathway.

- In the eukaryotes this pathway occurs in the mitochondria of the cell. It begins with the pyruvate which is formed as a result of breakdown of glucose in the cytoplasm (Via glycolysis or ED pathway)

- Under aerobic conditions i.e. in the presence of oxygen the pyruvate is converted to Acetyl CoA by the pyruvate dehydrogenase complex and then the Acetyl CoA continues with the Citric acid cycle in the mitochondria

- However, the prokaryotes lack cell organelles like mitochondria. So, in these organisms the Citric acid cycle occurs in the cell cytoplasm itself.

- All the enzymes utilized for the cycle are the same and all these enzymes in prokaryotes are present in the cell cytoplasm itself

- Not just in prokaryotes, the TCA cycle also occurs in the protists and in fungi.

- The TCA cycle involves various steps, it starts with the condensation of Acetyl CoA with Oxaloacetate (OAA) to form Citric acid.

- Since the first molecule that is formed in the pathway is the 3carbon citric acid molecule the pathway is given the name Citric acid cycle

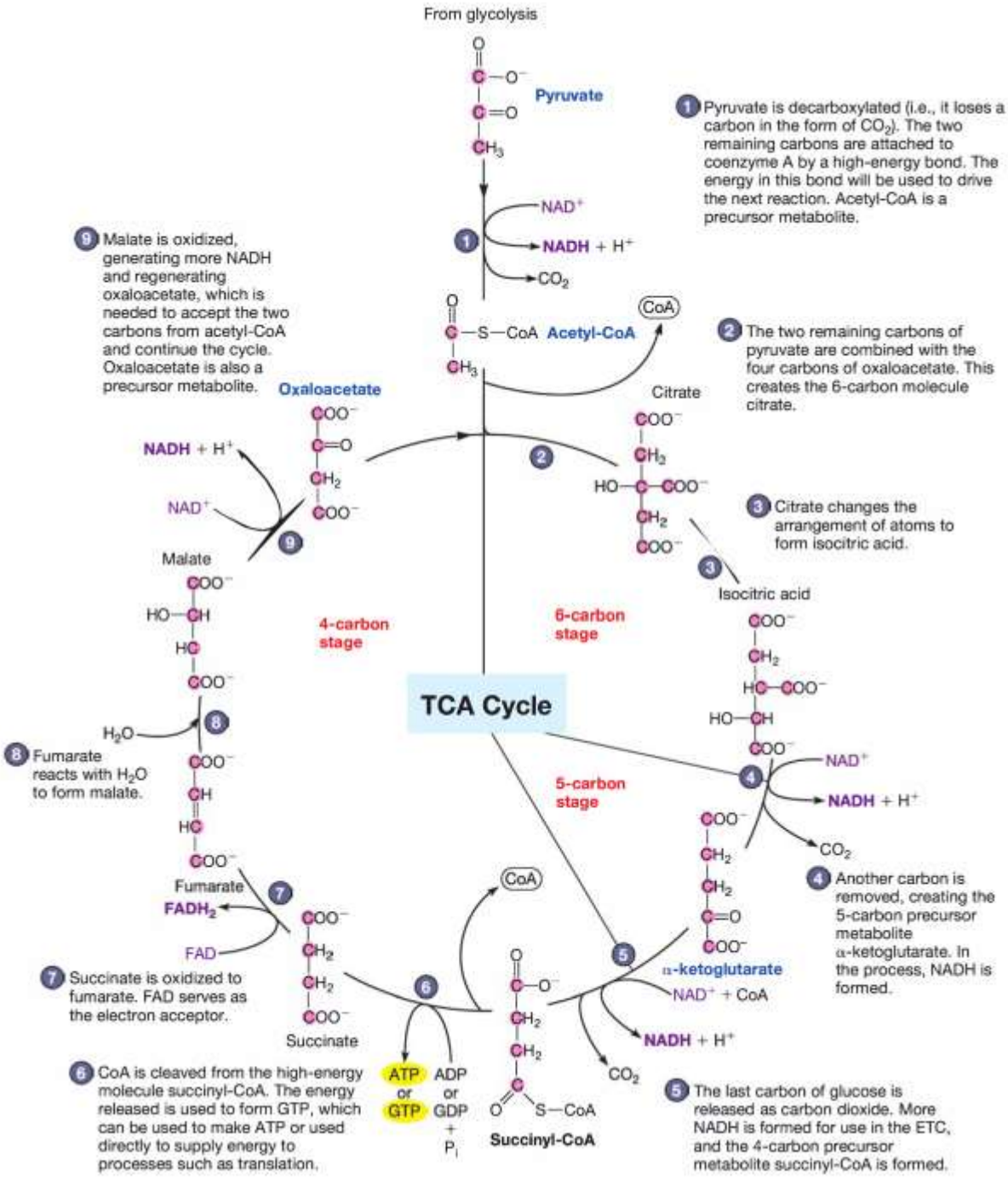
- The citrate then through a series of reactions (as shown in the image below) will replenish the Oxaloacetate back

- The carbons from the Acetyl CoA are lost in the form of carbon dioxide

- The overall pathways yields several NADH and $FADH_2$ reducing equivalents along with a GTP (Equivalent to ATP as represented in the equation)

- The entire cycle can be summarized in the following equation

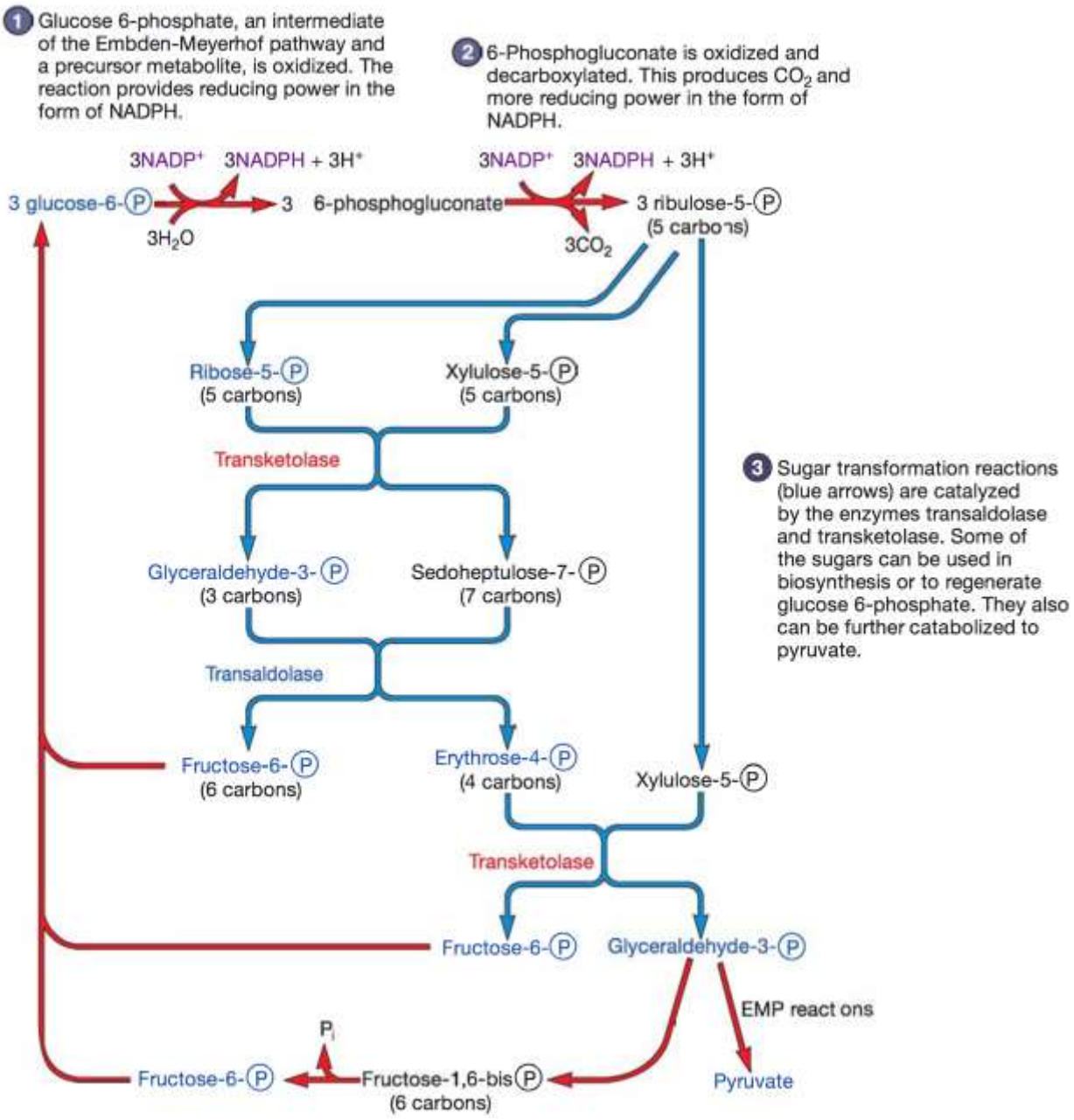
  $2 \text{ acetyl groups} + 6\ NAD^+ + 2\ FAD + 2\ ADP + 2\ P_i \rightarrow 4\ CO_2 + 6\ NADH + 6\ H^+ + 2\ FADH_2 + 2\ ATP$


From glycolysis
Pyruvate
1 Pyruvate is decarboxylated (i.e., it loses a carbon in the form of CO_2). The two remaining carbons are attached to coenzyme A by a high-energy bond. The energy in this bond will be used to drive the next reaction. Acetyl-CoA is a precursor metabolite.
NAD+
NADH + H+
CO2
CoA
Acetyl-CoA
2 The two remaining carbons of pyruvate are combined with the four carbons of oxaloacetate. This creates the 6-carbon molecule citrate.
Oxaloacetate
Citrate
NADH + H+
NAD+
9 Malate is oxidized, generating more NADH and regenerating oxaloacetate, which is needed to accept the two carbons from acetyl-CoA and continue the cycle. Oxaloacetate is also a precursor metabolite.
3 Citrate changes the arrangement of atoms to form isocitric acid.
Isocitric acid
Malate
4-carbon stage
6-carbon stage
TCA Cycle
5-carbon stage
H2O
8 Fumarate reacts with H2O to form malate.
NAD+
NADH + H+
CO2
4 Another carbon is removed, creating the 5-carbon precursor metabolite α-ketoglutarate. In the process, NADH is formed.
Fumarate
FADH2
FAD
7 Succinate is oxidized to fumarate. FAD serves as the electron acceptor.
CoA
α-ketoglutarate
NAD+ + CoA
NADH + H+
CO2
5 The last carbon of glucose is released as carbon dioxide. More NADH is formed for use in the ETC, and the 4-carbon precursor metabolite succinyl-CoA is formed.
Succinate
6 CoA is cleaved from the high-energy molecule succinyl-CoA. The energy released is used to form GTP, which can be used to make ATP or used directly to supply energy to processes such as translation.
ATP or GTP
ADP or GDP + Pi
Succinyl-CoA

PENTOSE PHOSPHATE PATHWAY (HEXOSE MONOPHOSPHATE PATHWAY OR HMP SHUNT)

- The HMP shunt occurs in the cell cytoplasm for 2 main reasons
    - Generation of pentose sugars especially ribose
    - Production of NADPH necessary for certain metabolic reactions (mainly lipid metabolism)
- The pathway begins with the oxidation of glucose-6-phosphate to 6-phosphogluconate
- 6-phosphogluconate is then oxidized to the 5 carbon ribulose-5-phosphate and CO_2
- Ribulose-5-phosphate is then converted to form a mixture of 7-carbon and 3-carbon sugar phosphates
- These reactions involve several transformations and the enzymes transketolases and transaldolases play an important role
- The overall outcome of this pathway can be summarized as follows:

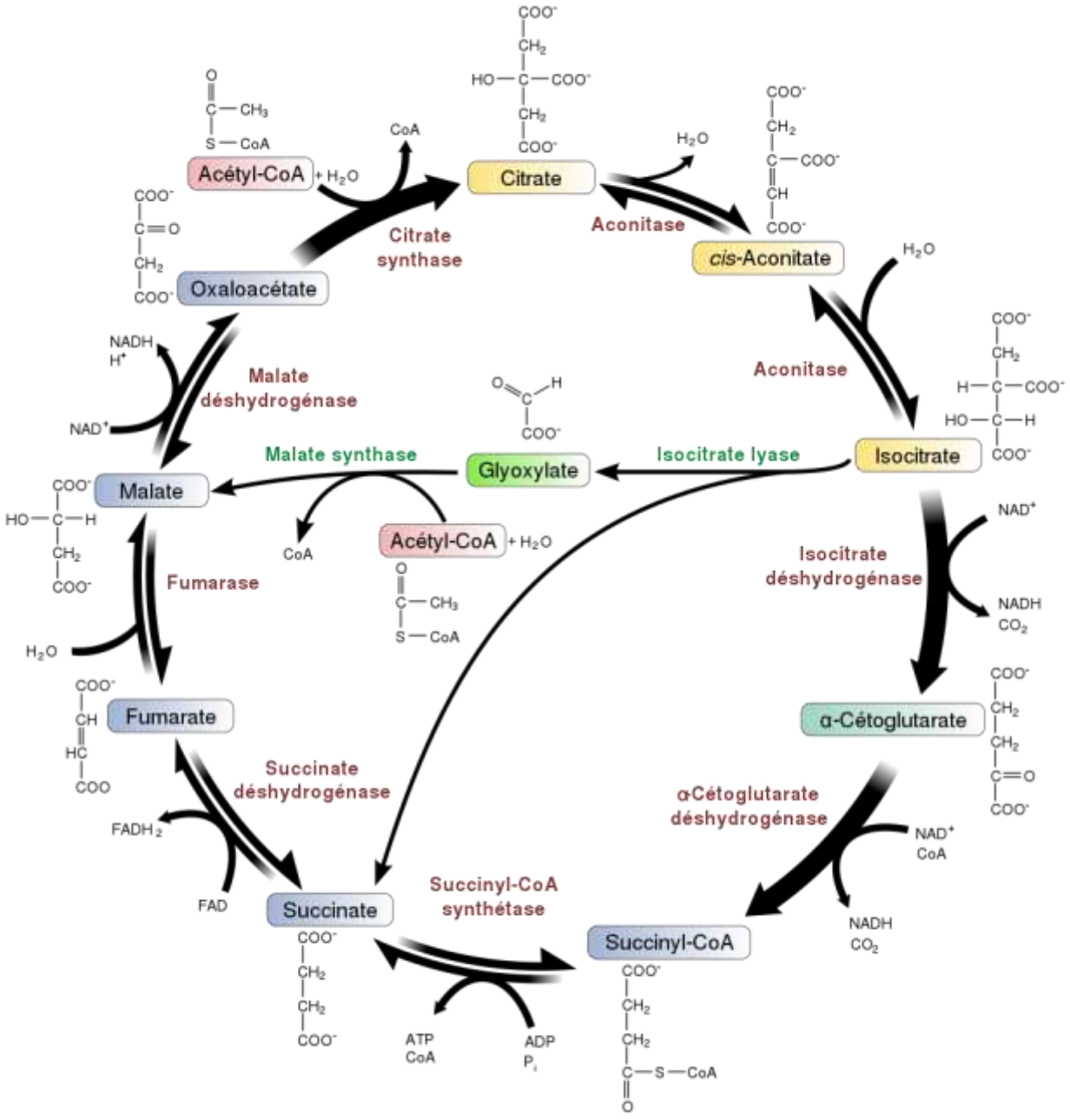
  3 glucose-6-phosphate + 6 $NADP^+$ + 3 H_2O → 2 fructose-6-phosphate + glyceraldehyde 3-phosphate + 3 CO_2 + 6 NADPH + 6 H^+
- In the process the pathway also produces erythrose 4-phospphate and ribose 5-phosphate that play important role acting as precursor metabolites for various amino acids, vitamins and nucleic acids



GLYOXYLATE CYCLE

- During the citric acid or the TCA cycle, the carbons in the Acetyl CoA molecules are lost in the form of carbon dioxide.

- In order to avoid this loss of carbon molecules in the form of carbon dioxide, some organisms use an alternate pathway
- For instance, in plants, certain invertebrates, and some microorganisms (including *E. coli* and yeast) the acetate will serve as both an energy-rich fuel and as a source of phosphoenolpyruvate for carbohydrate synthesis.
- The pathway involved here is called the Glyoxylate pathway
- In these organisms, enzymes of the **glyoxylate cycle** catalyze the net conversion of acetate to succinate (four carbon intermediate) of the citric acid cycle.
- Here, acetyl-CoA condenses with oxaloacetate to form citrate, which is then converted to isocitrate, exactly as in the citric acid cycle.
- In the next step, however, doe not involve isocitrate dehydrogenase but the cleavage of isocitrate by **isocitrate lyase,** forming succinate and **glyoxylate.**
- This glyoxylate then condenses with a second molecule of acetyl-CoA resulting in the formation of malate, a reaction catalyzed by **malate synthase.**
- Subsequently, Malate is oxidized to oxaloacetate, which can condense with another molecule of acetyl-CoA to begin another turn of the cycle.
- Each round of the glyoxylate cycle consumes two molecules of acetyl-CoA and produces one molecule of succinate, which is then available for biosynthetic purposes.



PHOTOSYNTHESIS

- Microorganisms cannot only derive energy from the oxidation of inorganic and organic compounds
- many can capture light energy and use it to synthesize ATP and NADH or NADPH .

- This process in which light energy is trapped and converted to **chemical energy** is called photosynthesis.
- Usually a photosynthetic organism reduces and incorporates CO2
- Photosynthesis is one of the most significant metabolic processes because all energy is derived from **solar energy**.
- It provides photosynthetic organisms with the ATP and NADPH necessary to synthesize the organic material for growth.
- In turn these organisms serve as the base of most food chains in the biosphere.
- Photosynthesis is also responsible for replenishing supply of O2
- Photosynthesis as a whole is divided into two parts: light and dark reactions.
- In the **light reactions,** light energy is trapped and converted to chemical energy.
- This energy is then used to reduce or fix CO2 and synthesize cell constituents in the **dark reactions.** The various reactions that occur during this dark reaction phase of the photosynthesis are a part of the **Calvin Cycle**

Diversity of Photosynthetic Organisms:

Eukaryotic Organisms	Prokaryotic Organisms
Higher plants	Cyanobacteria (blue-green algae)
Multicellular green, brown, and red algae	Green sulfur bacteria
Unicellular algae (e.g., euglenoids, dinoflagellates, diatoms)	Green nonsulfur bacteria
	Purple sulfur bacteria
	Purple nonsulfur bacteria
	Prochloron

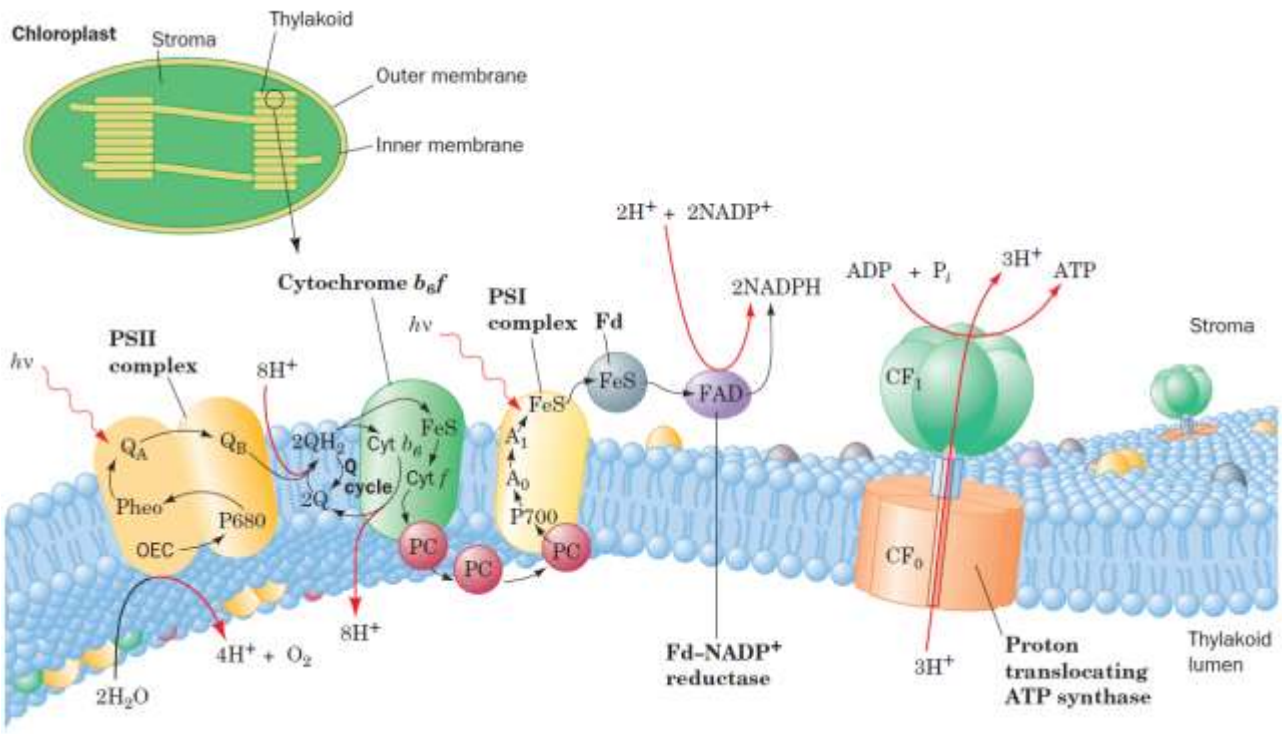
THE LIGHT REACTION IN EUKARYOTES AND CYANOBACTERIA

- All photosynthetic organisms have pigments for the absorption of light.
- The most important of these pigments are the chlorophylls.

- Chlorophylls are large planar rings composed of four substituted pyrrole rings with a magnesium atom coordinated to the four central nitrogen atoms.
- Several chlorophylls are found in eukaryotes, the two most important of which are **chlorophyll *a* and chlorophyll *b***.
- These two molecules differ slightly in their structure and spectral properties.
- When dissolved in acetone, chlorophyll *a* has a light absorption peak at 665 nm; the corresponding peak for chlorophyll *b* is at 645 nm.
- In addition to absorbing red light, chlorophylls also absorb blue light strongly (the second absorption peak for chlorophyll *a* is at 430 nm).
- Because chlorophylls absorb primarily in the red and blue ranges, green light is transmitted.
- Consequently, many photosynthetic organisms are green in color.
- The long hydrophobic tail attached to the chlorophyll ring aids in its attachment to membranes, the site of the light reactions.
- Other photosynthetic pigments also trap light energy.
- Most of these are the carotenoids, long molecules, usually yellowish, that possess conjugated double bond system.
- β-Carotene is present in *Prochloron* and most divisions of algae; fucoxanthin is found in diatoms, dinoflagellates, and brown algae (*Phaeophyta*).
- Red algae and cyanobacteria have photosynthetic pigments called **phycobiliproteins,** consisting of a protein with a tetrapyrrole attached.
- **Phycoerythrin** is a red pigment with a maximum absorption around 550 nm, and **phycocyanin** is blue (maximum absorption at 620 to 640 nm).
- **Carotenoids** and phycobiliproteins are often called accessory pigments because of their role in photosynthesis.
- **Chlorophylls** cannot absorb light energy effectively in the blue green through yellow range (about 470 to 630 nm)
- accessory pigments do absorb light and transfer the trapped energy to chlorophyll.
- In this way they make photosynthesis more efficient over a broader range of wavelengths.
- **Accessory pigments** also protect microorganisms from intense sunlight

- Chlorophylls and accessory pigments are assembled in highly organized arrays called **antennas**; whose purpose is to trap as many photons as possible.
- An antenna has about 300 chlorophyll molecules.
- Light energy is captured in an antenna and transferred from chlorophyll to chlorophyll
- Finally, it reaches a special reaction-center chlorophyll directly involved in photosynthetic electron transport.
- In eukaryotic cells and cyanobacteria, there are two kinds of antennas associated with two different **photosystems.**
- **Photosystem I** absorb longer wavelength light (>=680 nm) and funnels the energy to a special chlorophyll *a* molecule called P700.
- The term **P700** signifies that this molecule most effectively absorbs light at a wavelength of 700 nm.
- **Photosystem II** traps light at shorter wavelengths (<=680 nm) and transfers its energy to the special chlorophyll P680.
- When the photosystem I antenna transfers light energy to the reaction-center P700, it absorbs the energy and is excited and its reduction potential becomes very negative.
- It then donates its excited or high-energy electron to a chlorophyll *a* molecule (A) or an iron-sulfur protein.
- The electron is eventually transferred to **ferredoxin** and can then travel in the **cyclic pathway**, where the electron moves in a cyclic route through a series of electron carriers
- The pathway is termed cyclic because the electron from P700 returns to P700 after traveling through the photosynthetic electron transport chain.
- **PMF** is formed during cyclic electron transport in the region of cytochrome *b*6 and used to synthesize ATP.
- This process is called **cyclic photophosphorylation** because electrons travel in a cyclic pathway and ATP is formed.
- Electrons also can travel in a **noncyclic pathway** involving both photosystems.
- In the noncyclic route, P700 is excited and donates electrons to ferredoxin and reduced ferredoxin reduces NADP+ to NADPH.

- Because the electrons contributed to NADP+ cannot be used to reduce oxidized P700, photosystem II participation is required.
- It donates electrons to oxidized P700 and generates ATP in the process.
- The photosystem II antenna absorbs light energy and excites P680, which then reduces pheophytin *a*.
- **Pheophytin *a*** is **chlorophyll *a*** in which two hydrogen atoms have replaced the central magnesium.
- Electrons subsequently travel to Q (plastoquinone) and down the electron transport chain to P700.
- Oxidized P680 then obtains an electron from the oxidation of water to O_2.
- Thus, electrons flow from water all the way to NADP+ with the aid of energy from two photosystems, and ATP is synthesized by noncyclic photophosphorylation.
- It appears that one ATP and one NADPH are formed when two electrons travel through the noncyclic pathway.



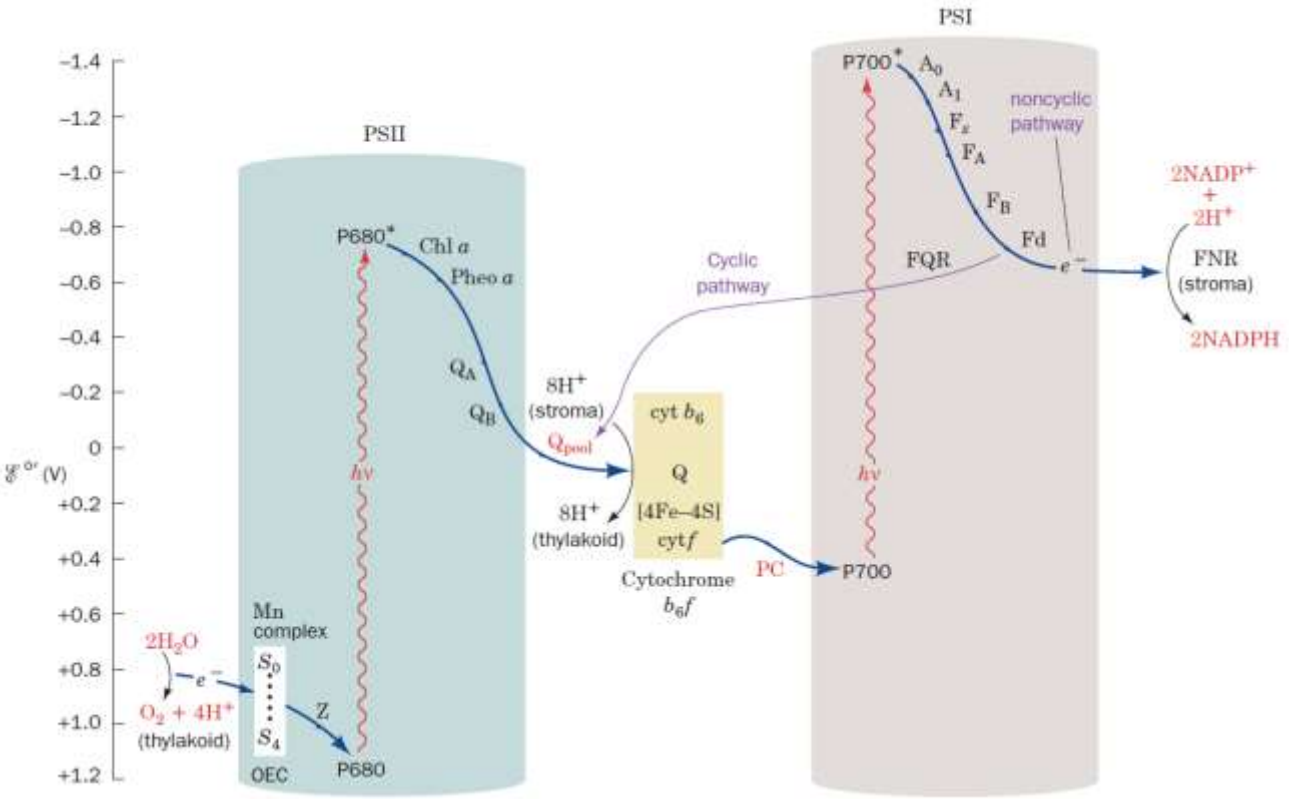
Schematic representation of the thylakoid membrane showing the components of its electron-transport chain. The system consists of three protein complexes: PSII, the cytochrome b6 f complex, and PSI, which are electrically "connected" by the diffusion of the electron carriers plastoquinol (Q) and plastocyanin (PC). Light-driven transport of electrons (black arrows) forms

NADPH motivates the transport of protons (red arrows) into the thylakoid space (Fd is ferredoxin). Additional protons are split off from water by the oxygen-evolving complex (OEC), yielding O_2. The resulting proton gradient powers the synthesis of ATP by the CF1CF0 proton-translocating ATP synthase [CF1 and CF0 are chloroplast (C) analogs of mitochondrial F1 and F0].The membrane also contains light-harvesting complexes whose component chlorophylls and other chromophores transfer their excitations to PSI and PSII.

- photosynthetic electron transport takes place within a membrane.
- Chloroplast **granal membranes** contain both photosystems and their antennas.
- Protons move to the **thylakoid** interior during photosynthetic electron transport and return to the **stroma** when ATP is formed.
- It is believed that stromal lamellae possess only photosystem I and are involved in cyclic photophosphorylation alone.
- In cyanobacteria, photosynthetic light reactions are also located in membranes.
- The dark reactions require three ATPs and two NADPHs to reduce one CO2 and use it to synthesize carbohydrate (CH_2O).

$$\mathbf{CO_2 + 3ATP + 2NADPH + 2H^+ + H_2O \rightarrow (CH_2O) + 3ADP + 3P_i + 2\ NADP^+}$$

- The noncyclic system generates one NADPH and one ATP per pair of electrons; therefore, four electrons passing through the system will produce two NADPHs and two ATPs.
- A total of 8 quanta of light energy (**4 quanta for each photosystem**) is needed to propel the four electrons from water to NADP+.
- Because the ratio of ATP to NADPH required for CO2 fixation is 3:2, at least one more ATP must be supplied.
- Cyclic photophosphorylation probably operates independently to generate the extra ATP.
- This requires absorption of another 2 to 4 quanta.
- It follows that around 10 to 12 quanta of light energy are needed to reduce and incorporate one molecule of CO2 during photosynthesis.



Detailed diagram of mechanism of photosynthesis. Electrons ejected from P680 by the absorption of photons are replaced with electrons abstracted from H2O by an Mn complex (OEC). Each ejected electron is passed through a chain of electron carriers to a pool of plastoquinone molecules (Q).The resulting plastoquinol, in turn, reduces the cytochrome b6 f particle (yellow box) that transfers electrons with the concomitant translocation of protons, via a Q cycle, into the thylakoid lumen. Cytochrome b6 f then transfers the electrons to plastocyanin (PC).The plastocyanin regenerates photo-oxidized P700.The electron ejected from P700, through the intermediacy of a chain of electron carriers is returned to the cytochrome b6 f complex in a cyclic process that only translocates protons into the thylakoid lumen.

- The Light Reaction in Green and Purple Bacteria Green and purple photosynthetic bacteria differ from cyanobacteria and eukaryotic photo-synthesizers
- In particular, green and purple bacteria do not use water as an electron source or produce O_2 photosynthetically—that is, they are **anoxygenic.**
- In contrast, cyanobacteria and eukaryotic photo-synthesizers are almost always oxygenic.
- NADPH is not directly produced in the photosynthetic light reaction of purple bacteria.
- Green bacteria can reduce NAD+ directly during the **light reaction.**

- To synthesize NADH and NADPH, green and purple bacteria must use electron donors that have more negative reduction potentials than water

- Finally, green and purple bacteria possess slightly different photosynthetic pigments, bacteriochlorophylls many with absorption maxima at longer wavelengths.

- **Bacteriochlorophylls** *a* and *b* have maxima in ether at 775 and 790 nm, respectively.

- This shift of absorption maxima into the infrared region better adapts these bacteria to their ecological niches.

- There are four groups of green and purple photosynthetic bacteria, each containing several genera: green sulfur bacteria (*Chlorobium*), green nonsulfur bacteria (*Chloroflexus*), purple sulfur bacteria (*Chromatium*), and purple nonsulfur bacteria (*Rhodospirillum, Rhodopseudomonas*).

- Many differences found in green and purple bacteria are due to their lack of photosystem II and inability to use water as an electron donor in noncyclic electron transport.

- Without photosystem II they cannot produce O_2 from H_2O photosynthetically and are restricted to **cyclic photophosphorylation.**

- Indeed, almost all purple and green sulfur bacteria are **strict anaerobes.**

- When the special reaction-center chlorophyll P870 is excited, it donates an electron to bacteriopheophytin.

- Electrons then flow to quinones and through an electron transport chain back to P870 while driving ATP synthesis.

- the purple bacteria have a photosynthetic apparatus similar to photosystem II, whereas the green sulfur bacteria have a system similar to photosystem I.

- Green and purple bacteria face a further problem because they also require NADH or NADPH for CO_2 incorporation.

- They may synthesize NADH in at least three ways.

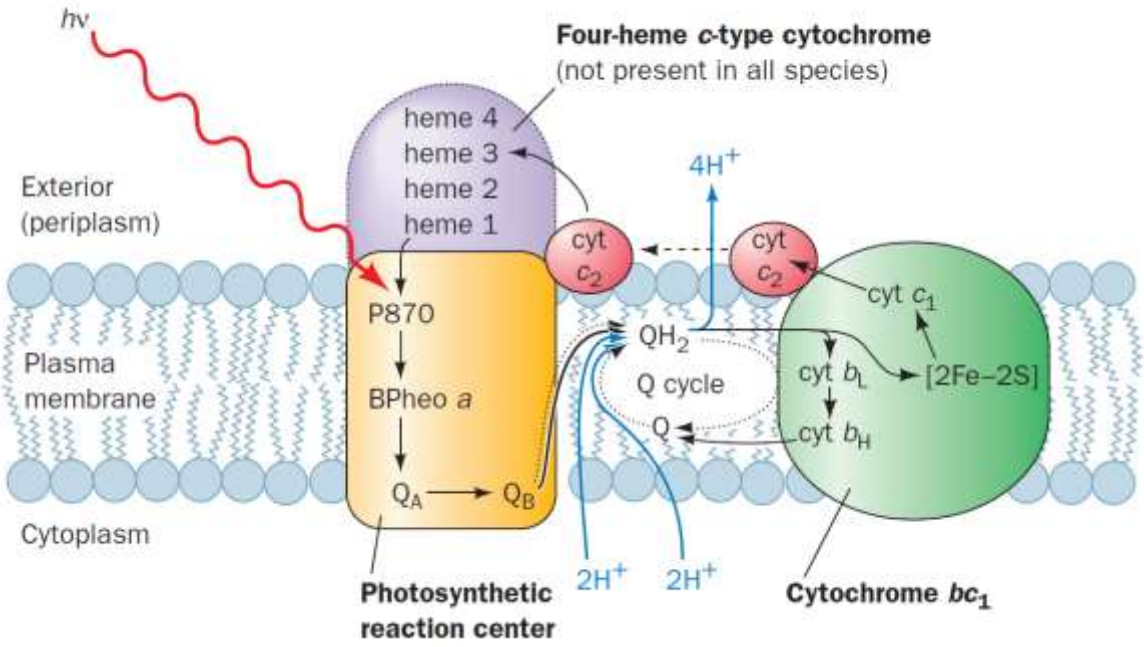
- If they are growing in the presence of hydrogen gas, the hydrogen can be used directly to produce NADH.

- Many photosynthetic purple bacteria use proton motive force to reverse the flow of electrons and move them from inorganic or organic donors to NAD+.

- Green sulfur bacteria such as *Chlorobium* appear to carry out a simple form of noncyclic photosynthetic electron flow to reduce NAD+.

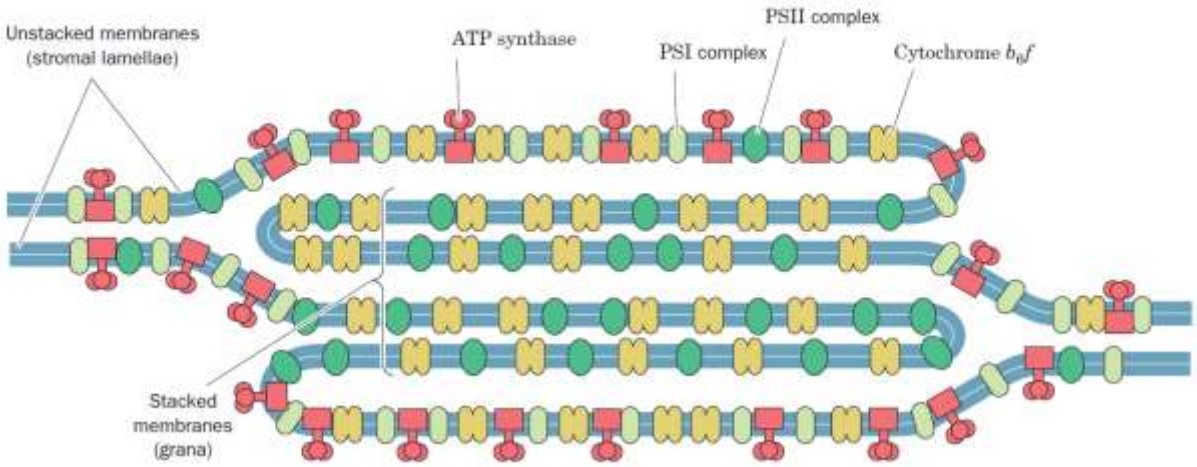
Properties of Microbial Photosynthetic Systems:

Property	Eukaryotes	Cyanobacteria	Green and Purple Bacteria
Photosynthetic pigment	Chlorophyll *a*	Chlorophyll *a*	Bacteriochlorophyll
Photosystem II	Present	Present	Absent
Photosynthetic electron donors	H_2O	H_2O	H_2, H_2S, S, organic matter
O_2 production pattern	Oxygenic	Oxygenic	Anoxygenic
Primary products of energy conversion	ATP + NADPH	ATP + NADPH	ATP
Carbon source	CO_2	CO_2	Organic and/or CO_2



Photosynthetic electron-transport system of purple photosynthetic bacteria. Schematic diagram indicating the arrangement of the system components in the bacterial plasma membrane and the flows of electrons (black arrows) and protons (blue arrows) that photon absorption promotes through them. The system contains two protein complexes, the RC and cytochrome bc1.Two electrons liberated from the special pair, here P870 (as in Rb. sphaeroides), by the consecutive absorption of two photons are taken up by ubiquinone (QB) together with two

protons from the cytoplasm to yield ubiquinol (QH_2).The QH_2 is released from the RC and diffuses (dotted arrows) through the membrane to cytochrome bc1, which, in a two-electron reaction, oxidizes it to ubiquinone with the concomitant liberation of its two protons to the external medium. One of the two electrons is passed, via the [2Fe–2S] cluster and cytochrome c1, to cytochrome c2, a peripheral membrane protein that then diffuses across the external surface of the membrane so as to return the electron to P870 of the RC.



Segregation of PSI and PSII. The distribution of photosynthetic protein complexes between the stacked (grana) and the unstacked (stroma exposed) regions of the thylakoid membrane is shown.

- The light reaction ultimately produces the ATP and the NADPH required for the synthesis of the sugars from the carbon dioxide in the subsequent steps i.e. the dark reaction or the Calvin Cycle

CALVIN CYCLE

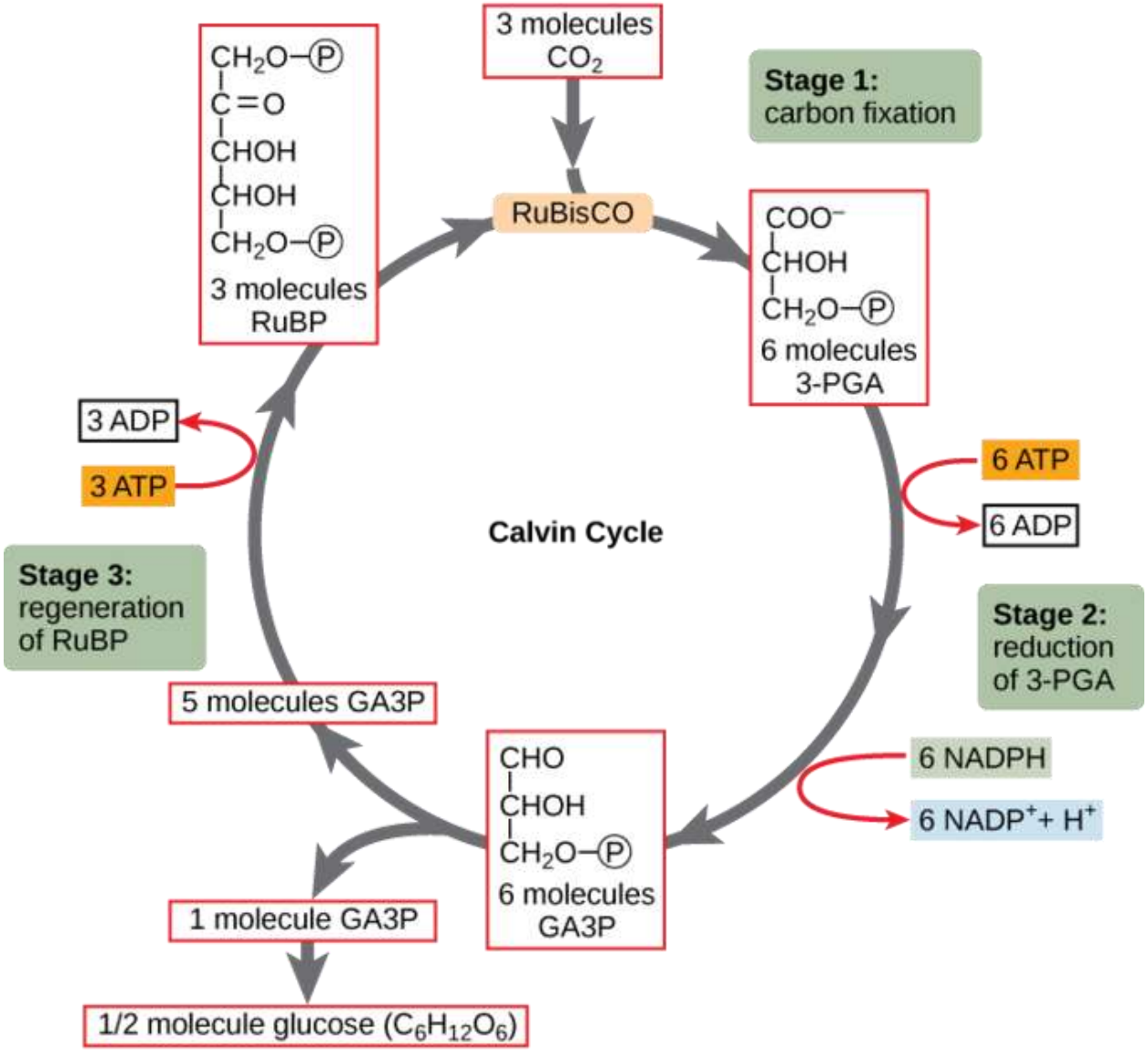
- In Thiobacilli, Cyanobacteria and nitrifying bacteria the Calvin cycle occurs in the inclusions called the **Carboxysomes**

- The Calvin cycle is divided into 3 phases: carboxylation phase, reduction phase, and regeneration phase.

    - **Carboxylation phase or Carbon fixation phase**: In this phase, Ribulose 1,5-bisphosphate carboxylase/oxygenase (RubisCO) enzyme catalyzes the addition of CO_2 to the 5-carbon molecule ribulose 1,5-bisphosphate (RuBP), forming a 6-carbon intermediate which is unstable and rapidly splits into two molecules of 3-phosphoglycerate (PGA).

- **Reduction phase:** PGA is reduced to glyceraldehyde 3-phosphate by two reactions. The enzyme glyceraldehyde 3-phosphate dehydrogenase uses NADPH

- **Regeneration phase:** As the name suggests, RuBP is regenerated, so that the cycle can repeat. Additionally, this phase produces carbohydrates such as glyceraldehyde 3-phosphate, fructose 6-phosphate, and glucose 6-phosphate, all of which are precursor metabolites.



- The overall pathway can be summarized in the equation below

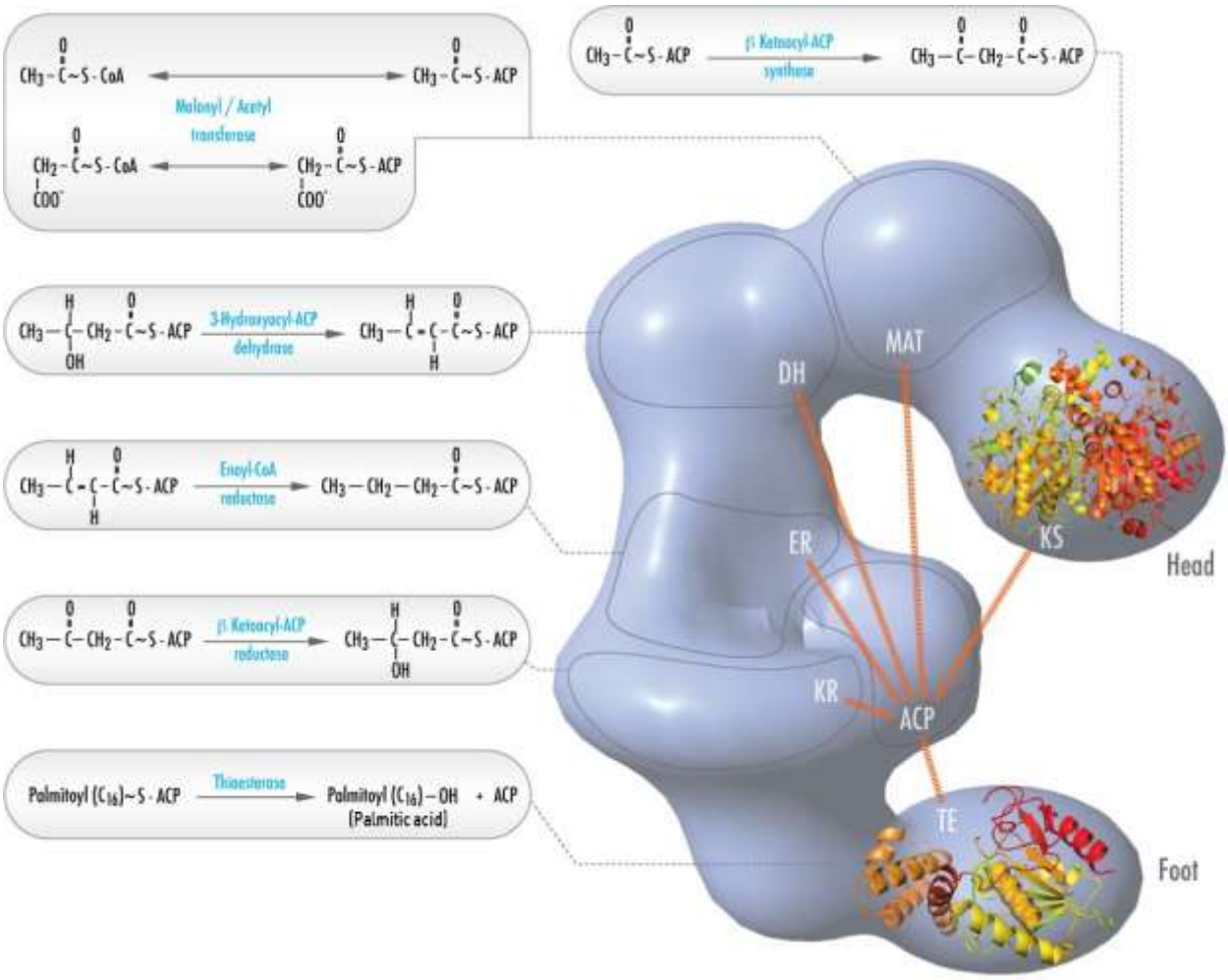
  $6\ CO_2 + 12\ NADPH + 12\ H^+ + 12\ H_2O + 18\ ATP \rightarrow Glucose + 12\ NADP^+ + 18\ ADP + 18\ Pi$

LIPID METABOLISM

- Lipids are the most important components of all the membranes. So, all types of cells need to synthesize the lipids
- Most bacterial and eukaryotic lipids contain fatty acids or their derivatives

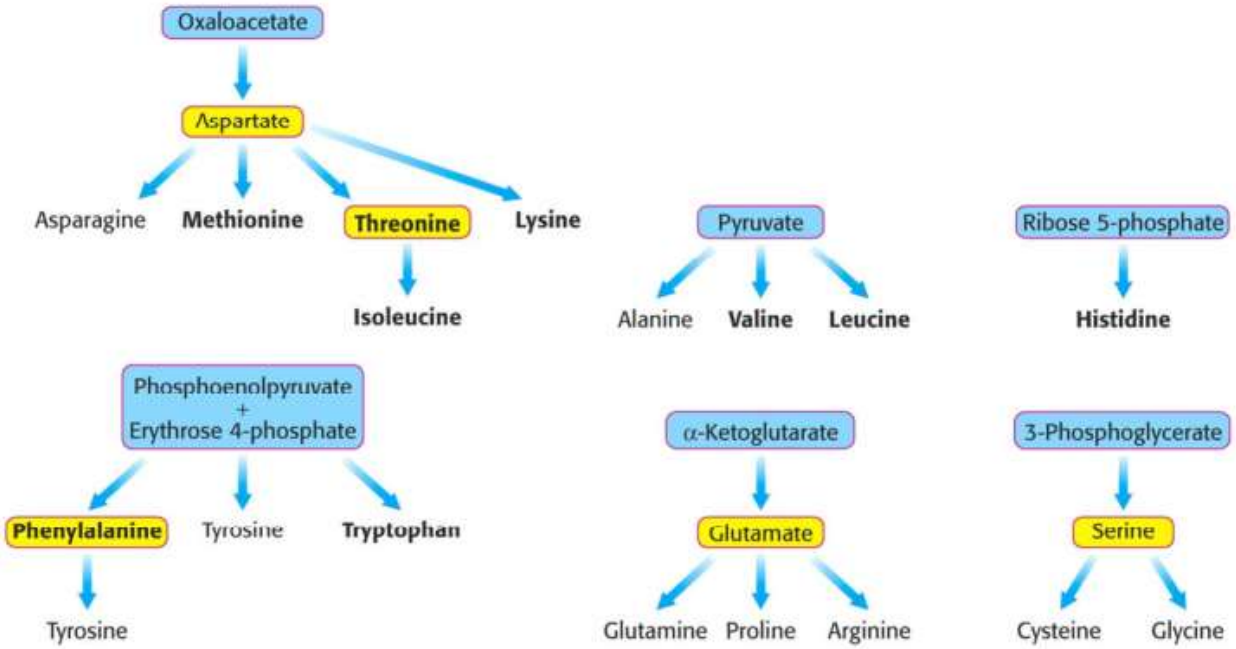
FATTY ACID SYNTHESIS

- The enzyme fatty acid synthase complex (shown in the image below with a head and foot portion) is a complex of many different enzymes and it catalyzes the Fatty acid synthesis
- The synthesis starts with acetyl-CoA and malonyl-CoA as the substrates and NADPH as the electron donor.
- Acetyl CoA is converted to Malonyl-CoA by a carboxylation reaction which is catalyzed by the enzyme acetyl-CoA Carboxylase.
- In the next step CoA is removed from both acetyl CoA and Malonyl CoA and replaced with the Acyl Carrier Protein (ACP) to from acetyl-ACP and Malonyl-ACP respectively.
- Then the malonyl-ACP reacts with the acetyl-ACP to form a fatty acyl-ACP two carbons longer
- Malonyl-ACP releases the third carbon (that was added initially) in the form of carbon dioxide.
- In the second stage of synthesis, the beta-keto group that is the result of the initial condensation reaction is removed in a three-step process involving two reductions and a dehydration.
- In the image shown below, the various steps involved in the synthesis of the fatty acid, palmitic acid can be seen in the image below



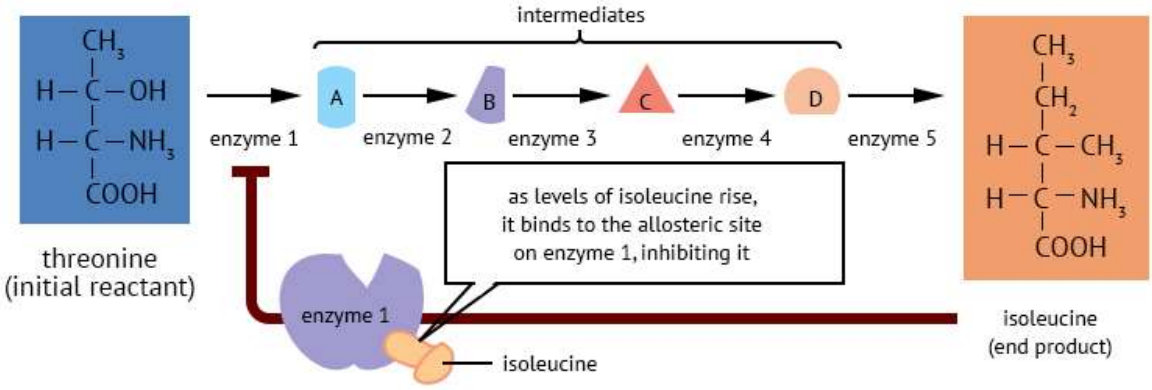
AMINO METABOLISM (AMINO ACID SYNTHESIS)

- Amino acids are the building blocks of the proteins
- They are made from the various intermediates of the TCA cycle and the other metabolic pathways
- Unlike Eukaryotes, most of the prokaryotes have the ability to synthesize most of the amino acids on their own using the various intermediates
- The 20 standard amino acids are synthesized by various complex pathways and the image below gives a summary of the various precursor molecules used for their synthesis



REGULATION OF AMINO ACID SYNTHESIS

- The rate at which amino acids are synthesized depends mainly on the amounts of the biosynthetic enzymes and on their activities.
- The amino acid biosynthesis is mainly regulated by the feedback inhibition mechanism in which the product when made in excess will bind to the enzymes synthesizing them and inhibit their activity
- Usually the enzyme catalyzing the committed step in the pathway is regulated by feedback inhibition
- Take the example of the synthesis of isoleucine from threonine. It involves several reactions and several enzymes are involved in the pathway.
- When the cell gets enough amounts of the isoleucine, the isoleucine binds to the allosteric site on the enzyme that catalyzes the first step of its biosynthetic pathway.
- Binding to the allosteric site on the enzyme alters the conformation of the enzyme leading to the inhibition of the enzyme ultimately blocking the subsequent steps in the pathway.



ELECTRON TRANSPORT CHAIN

- The various reducing equivalents i.e. NADH and the $FADH_2$ that are generated during the glycolytic, TCA and the other metabolic pathways need to donate their electrons and get back to their original NAD and FADH state so that they can participate in more metabolic reactions

- So, these molecules done their electrons to some components that are located within the mitochondrial membrane in the case of Eukaryotes or in the plasma membrane in the case of bacteria and Archaea

- In the case of the Gram-Negative bacteria, these Electron Chain Components can be found in the periplasmic space and the outer membrane

- The amino acid biosynthesis is mainly regulated by the feedback inhibition mechanism in which the product when made in excess will bind to the enzymes synthesizing them and inhibit their activity